



6829 - Understanding the rich molecular spectra observed with NIRspec and MIRI/MRS toward protostellar outflows

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Spectroscopic observations of protstellar outflows and jets with JWST reveal an extraordinary wealth of molecular lines. Carefully interpreted with the use of detailed molecular excitation models, they will provide unique physical and chemical information about the local environment, such as the gas temperature, gas density, composition and ambient UV field. The AR theory proposal proposed here has two key elements:

(1) We will model the excitation of H₂, CO, H₂O, CH₄, CO₂, and C₂H₂ bands that have already been observed by JWST, focusing primarily but not exclusively on five sources targeted in the "Investigating Protostellar Accretion" (IPA) program (PID 1802). The expected outcome of the analysis will be more reliable estimates of the molecular abundances, gas temperature and density (and how they vary spatially). These parameters are key for understanding the nature of protostellar outflows; and

(2) we will create a set of tools of general applicability to the target molecules in dense astrophysical environments. These will be made available to the community and will prove useful for the interpretation of molecular emissions from a wide range of astrophysical sources that have been/will be targeted in previous/future JWST observations.

OBSERVING DESCRIPTION